

Naturally-Colored Translation from LTL to Parity and COCOA

Rüdiger Ehlers and *Ayrat Khalimov*, TU Clausthal

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- Each L^i is a coBuchi language – has a canonical coBuchi automaton^[AK22].
- COCOA = the canonical chain of co-Buchi automata.

Motivation for this work

- The only known algorithm to construct COCOA from LTL is the route $LTL \rightarrow DPA \rightarrow COCOA$. The main step $LTL \rightarrow DPA$ ignores natural colors.
- *Is there natural-colors inspired translation $LTL \rightarrow COCOA$ that avoids detours to Safra and Esparza et al.?*
- *Does the natural-color decomposition enable a new route $LTL \rightarrow DPA$?*

This paper

Direct translations using natural-color decomposition:

$$\begin{array}{l} \text{LDL} \rightarrow \text{WAA} \rightarrow \left\{ \begin{array}{l} \text{SLTM} \\ \text{NBA}_{\varphi} \\ \text{NBA}_{\neg\varphi} \end{array} \right\} \rightarrow \begin{array}{c} \text{Floating Automata} \\ \text{for } L^1, \dots, L^k \end{array} \rightarrow \left\{ \begin{array}{l} \text{COCOA } (c^1, \dots, c^k) \\ \text{DPA} \\ \text{LDBA} \end{array} \right\} \\ (\text{LTL}) \end{array}$$

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Plan:

- definition of natural colors and COCOA
- SLTM, Floating Automata (DfA)
- incremental construction of DfA for level- ℓ L^ℓ
- from DfA to COCOA and DPA

Natural colors and COCOA

Fix a language L , a word $w = w_0 w_1 \dots$, and an infinite set $J = j_1 < j_2 < \dots$.
A word $w' = w_0 \dots w_{j_1-1} u_1 w_{j_1} \dots w_{j_2-1} u_2 w_{j_2} \dots$, s.t. every u_i preserves the suffix language of $w_0 \dots w_{j_i-1}$ wrt. L is *suffix-language-invariant injection*.

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Example: $L = L(\text{FG}a)$, $a^\omega \in L$, $J = \mathbb{N}$, injecting $\bar{a}$ yields $(\bar{a}a)^\omega \notin L(\text{FG}a)$.

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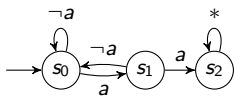
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The SLTM (right-congruence automata)

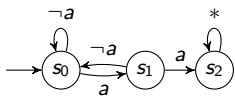


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State s_0 encodes $L(\varphi)$,

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Lemma^[AF24]. *Given a weak alternating automaton with n states, one can construct the SLTM with $2^{2^{O(n)}}$ states in time $2^{2^{O(n)}}$.*

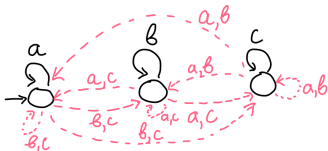
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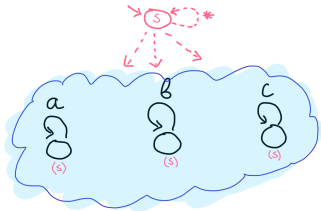
The HD-NCA for $FG(a) \vee FG(b) \vee FG(c)$:



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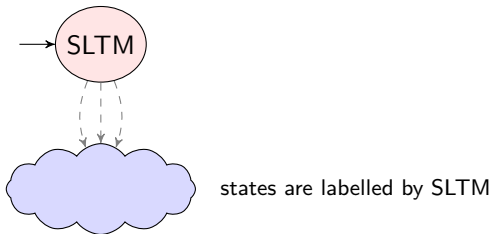
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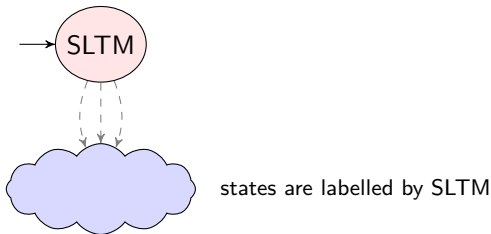
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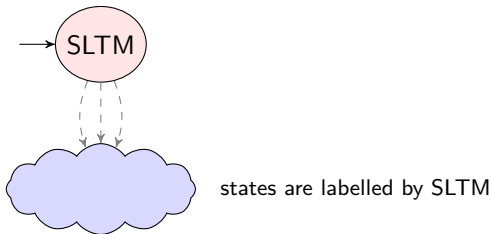
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Lemma. *Every deterministic floating automaton can be translated into an equivalent HD-NCA in polynomial time.*

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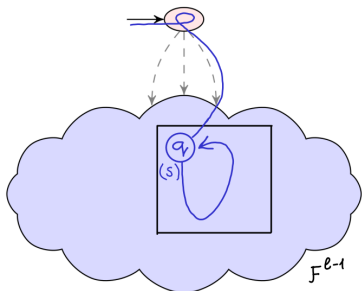
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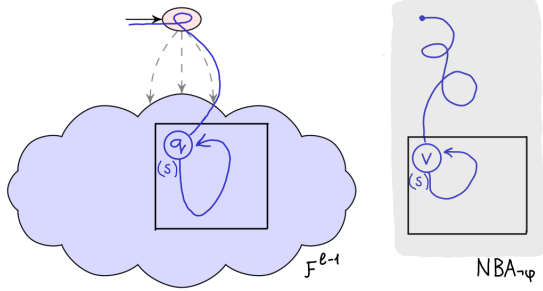


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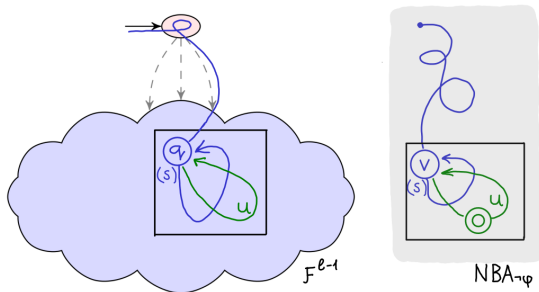


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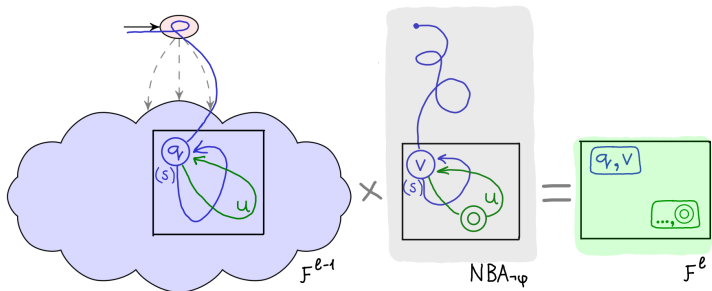


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Theorem. *Given a weak alternating automaton with n states, the procedure constructs the COCOA with $2^{2^{O(n)}}$ states in time $2^{2^{O(n)}}$. Each HD-NCA/DfA has size at most $2^{2^{O(n)}}$ and this is tight for full LTL.*

Translating LDL into DPA

Variant 1: DfA $\rightarrow$ DCA then product:

- Determinize DfAs $(\mathcal{F}^1, \dots, \mathcal{F}^k)$ into DCAs $(\mathcal{C}^1, \dots, \mathcal{C}^k)$.
- Build the product of DCAs: the parity color of the joint transition is the index-1 of the minimal DCA that takes a rejecting transition.
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Variant 2: eager product of DfA:

- Challenge: DfAs are nondeterministic thanks to the jump.
- *Eager* resolution of the jump moment.
- 2EXP size bound but (probably) loses the acceptance with the nat. color.

Theorem. *Given a weak alternating automaton with n states, the procedure LDLtoDPA constructs DPA with $2^{2^{O(n)}}$ states in time $2^{2^{O(n)}}$.*

Conclusion

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(LTL)

- Strengths: handles WAA $\Rightarrow$ handles LDL, relatively easy to understand, many parts are efficiently implementable (graph based), modular
- Weaknesses: SLTM construction, too many abstractions (loss of semantic connection with the original LDL)